

Conversions

1. Rapid conversions
2. Substitution method
3. Successive divisions and multiplication

1. **Rapid conversions:** conversions between bases which are powers of 2.

1.1 Conversion from the source base $p=2^k$, $p\{4=2^2, 8=2^3, 16=2^4\}$ into the destination base 2

Rule:

Each digit from the source number in base $p=2^k$, the integer part and the fractional one, will be replaced by the corresponding group of k binary digits (adding if it is necessary insignificant zeros to the left).

1.2. Conversion from base 2 into the destination base $q=2^k$, $q\{4=2^2, 8=2^3, 16=2^4\}$

Rules:

- for the **integer/fractional part**: from **right/left** to **left/right** (relative to the decimal point) make groups of k binary digits (eventually we add to the **left/right** insignificant zeros to have a complete group);
- the groups will be replaced by the corresponding digits in base $q=2^k$.

Example 1:

$$(8) = ? (2) = ? (16)$$
$$5327,321(8) = 101\ 011\ 010\ 111,011\ 010\ 001(2) =$$
$$1010\ 1101\ 0111,0110\ 1000\ 1000(2) = AD7,688(16)$$

$$5(8) = 5 = 4 + 1 = 2^2 + 2^0 = 101(2)$$
$$D(16) = 13 = 2^3 + 2^2 + 2^0 = 1101(2)$$
$$115 = 64 + 32 + 16 + 2 + 1 = 2^6 + 2^5 + 2^4 + 2^1 + 2^0 = 1110011(2)$$

Example 2:

$$03\ 32\ 10,32\ 30(4) = 3E4,EC(16)$$
$$32(4) = 2 * 4^0 + 3 * 4^1 = 2 + 12 = 14 = E(16)$$

Example 3:

$$6BEA, BEC(16) = 12\ 23\ 32\ 22, 23\ 32\ 30(4)$$
$$B(16) = 11 = 2 * 4^1 + 3 * 4^0 = 23(4)$$
$$E(16) = 14 = 3 * 4 + 2 = 32(4)$$

2. Substitution method

- **calculations in the destination base**
- it is recommended for $b < h$, b (source base), h (destination base)

Steps:

- all the digits from the source representation are converted into the destination base:
- the base b is converted into base h :
- we calculate in base h the sum of the positional values

Example 4:with a precision of 2 digits at the fractional part in the destination representation

1432,31 ₍₅₎ = ? , ₍₈₎

$$1(5) = 1(8), \quad 4(5) = 4(8), \quad 3(5) = 3(8), \quad 2(5) = 2(8), \quad 5 = 5(8)$$

$$1432,31(5) = 1(8) * 5(8) \wedge 3 + 4(8) * 5(8) \wedge 2 + 3(8) * 5(8) \wedge 1 + 2(8) * 5(8) \wedge 0 +$$

$$+ 3(8) * 5(8) \wedge (-1) + 1 * 5(8) \wedge (-2) = 175(8) + 144(8) + 17(8) + 2(8) + 0,46(8) + 0,02(8) =$$

$$= 362,50(8)$$

!!! Calculations in base 8!!!

[illegible]

$$5(8)^2 = 31(8)$$

$5 \times 5 = 25$, $25 \bmod 8 = 1$, carry = 3

$$5(8)^3 = 175(8)$$

$3 \cdot 5 = 15$, $15 \bmod 8 = 7$, carry = 1

$4 \times 3 = 12, 12 \bmod 8 = 4$, carry 1

$$3(8)*5(8)=17(8)$$

3,00 (8)	: 5(8)	1,00(8)	5(8)	0,14(8)	5(8)
/	0,46 (8)	/	0,14	/	0,02
30		10		1	
/		/		/	
40		30		14	
2		4		2	

$$30(8) = 3 * 8 + 0 = 24, 24 \div 5 = 4, 24 \bmod 5 = 4$$

$$40(8) = 4 * 8 + 0 = 32, 32 \div 5 = 6, 32 \bmod 5 = 2$$

$$10(8) = 1 * 8 = 8 = 8, 8 \text{ div } 5 = 1, 8 \text{ mod } 5 = 3$$

$$30(8) = 3 * 8 = 24, 24 \text{ div } 5 = 4, 24 \text{ mod } 5 = 4$$

$$14(8) = 8 + 4 = 12, 12 \operatorname{div} 5 = 2, 12 \bmod 5 = 2$$

carries	1	1	0		1		+(16)
	1	5	7	,	0	0	
		C	4		0	0	
		2	3	,	0	0	
			0	,	6	D	
			0	,	1	F	
			2	,	0	0	
	2	4	0	,	8	C	

1452, 36 (7) = 240, 8C (16)

3. The method of successive divisions/multiplications

- it is recommended for $h < b$, b –source base and h - destination base.
- calculations in the source base

Integer part: successive divisions by the destination base (h) are performed

- the process of successive divisions ends when 0 is obtained as quotient.
- the remainders, in the reverse order, are the digits of the new representation in base q .

Fractional part: successive multiplications by the destination base (h) are performed

- the fractional part is multiplied by b obtaining a number with an integer part and a fractional one;
- we continue with the multiplication of this new fractional part,...
- the process of the successive multiplications continues until one of the following conditions is satisfied:
 - a) the fractional part becomes 0;
 - b) an established number of digits of the fractional part were calculated;
 - c) periodicity is obtained.
- the integer parts, in the order of obtaining them during the multiplications process, are the digits of the fractional part in the destination representation.

Example 6: with a precision of 3 digits at the fractional part in the destination representation

!!calculations in base 8

$$362,50_{(8)} = ?_{(5)}$$

Conversion of the integer part

362 (8)	:5(8)	60(8)	:5(8)	11(8)	:5(8)	1(8)	:5(8)
36	60	10	11	4	1	1	0
02		3					
2							

$$36(8) = 6 + 3 * 8 = 6 + 24 = 30, 30 / 5 = 6, 30 \% 5 = 0$$

$$10(8) = 8, 8 / 5 = 1, 8 \% 5 = 3$$

$$11(8) = 1 + 8 = 9, 9 / 5 = 1, 9 \% 5 = 4$$

Calculations:

Conversion of the fractional part : $0,5(8)=0,(30)_{(5)}$

c	3													
	0,	5	*				*				*			
		5												
	3,	1												

$$0,5(8)*5(8)=3,1(8)$$

$$25 \text{ div } 8 = 3, 25 \text{ mod } 8 = 1$$

$$0,1(8) * 5(8)=0,5(8)$$

$$362,50_{(8)} = \sim 1432,(30)_{(5)}$$

Example 7: with a precision of 3 digits at the fractional part in the destination representation

!!! Calculations in base 16

$$240, 8C_{(16)} = ?_{(7)}$$

Conversion of the integer part :

240(16)	:7(16)	52(16)	:7(16)	B(16)	:7(16)	1(16)	:7(16)
24	52	5	B	4	1	1	0

10			
2			

$$24(16) = 4 + 2 * 16 = 4 + 32 = 36, \quad 36 \bmod 7 = 1, \quad 36 \operatorname{div} 7 = 5$$

$$10(16) = 16, \quad 16 \bmod 7 = 2, \quad 16 \operatorname{div} 7 = 2$$

$$52(16) = 2 + 5 * 16 = 2 + 80 = 82, \quad 82 \bmod 7 = 5, \quad 82 \operatorname{div} 7 = 11 = B(16)$$

Conversion of the fractional part $0,8C(16) = 0, \text{35} \quad (7)$

c	3	5	0			55	1	0						
	0,	8	C	*		0,	D	4	*				*	
			7					7						
	3,	D	4			5,5,	CC	C						

$$0,8C(16) * 7(16) = \text{3,D4}(16)$$

$$C(16) * 7(16) = 12 * 7 = 84, \quad 84 / 16 = 5, \quad 84 \bmod 16 = 4$$

$$8 * 7 + 5 = 61, \quad 61 / 16 = 3, \quad 61 \% 16 = 13 = D(16)$$

$$\text{0,D4}(16) * 7(16) = \text{5,CC}(16)$$

$$4 * 7 = 28, \quad 28 / 16 = 1, \quad 28 \% 16 = 12 = C(16)$$

$$D(16) * 7(16) + 1(16) = 13 * 7 + 1 = 92, \quad 92 / 16 = 5, \quad \% 16 = 12 = C(16) \quad 92 \% 16 = C(16)$$

$$\text{240, 8C} (16) \approx \text{1452, 35} (7)$$